



CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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0620/42

February/March 2023

1 hour 15 minutes

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [].
- The Periodic Table is printed in the question paper.

This document has **12** pages.

1 This question is about gases found in clean, dry air and gases found in polluted air.

(a) Name **one** gas found in clean, dry air which contributes to global warming.

..... [1]

(b) State the percentage of nitrogen in clean, dry air.

..... [1]

(c) Name the substance used to remove sulfur dioxide in flue gas desulfurisation.

..... [1]

(d) Nitrogen dioxide, NO_2 , is formed in car engines.

Name the equipment in a car exhaust used to remove the NO_2 formed in car engines.

..... [1]

(e) All gases diffuse.

(i) Choose from the list of formulae the gas which diffuses most quickly.

Draw a circle around your answer.

CO CO₂ CH₄ NO₂ SO₂

[1]

(ii) Explain your answer to (i).

..... [1]

(f) State **one** adverse effect of carbon monoxide on human health.

..... [1]

(g) Carbon dioxide, CO_2 , is a reactant in photosynthesis.

Name the **two** products of photosynthesis.

..... and [2]

(h) Complete the dot-and-cross diagram in Fig. 1.1 for a molecule of CO_2 .

Show outer shell electrons only.

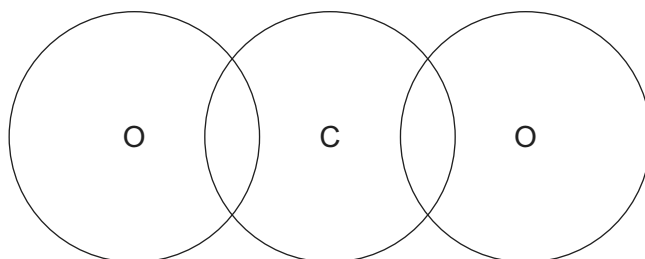


Fig. 1.1

[2]

[Total: 11]

2 Lithium, sodium and potassium are Group I elements.

(a) Name the type of bonding in these elements.

..... [1]

(b) Sodium reacts with cold water to form hydrogen gas and a solution of a strong alkali.

(i) State the test for hydrogen gas.

test

positive result

[1]

(ii) Suggest the pH of a solution of a strong alkali.

pH = [1]

(iii) Name a substance which can be used to confirm the pH of a solution of a strong alkali.

..... [1]

(iv) Write the symbol equation for the reaction between sodium and cold water.

Include state symbols.

..... [3]

(c) Lithium has two naturally occurring types of atoms, ${}^6\text{Li}$ and ${}^7\text{Li}$.

(i) State the name given to atoms of the same element with different nucleon numbers.

..... [1]

(ii) Complete Table 2.1 to show the number of protons, neutrons and electrons in the atom and ion of lithium shown.

Table 2.1

	${}^6\text{Li}$	${}^7\text{Li}^+$
protons		
neutrons		
electrons		

[3]

- (iii) Table 2.2 shows the relative abundance of the two naturally occurring atoms of lithium.

Table 2.2

atom	${}^6\text{Li}$	${}^7\text{Li}$
relative abundance	10%	90%

Calculate the relative atomic mass of lithium to **one** decimal place.

relative atomic mass = [2]

- (d) Potassium oxide, K_2O , is an ionic compound.

Complete Fig. 2.1 to show the electronic configurations of the ions in potassium oxide.
Show the charges on the ions.

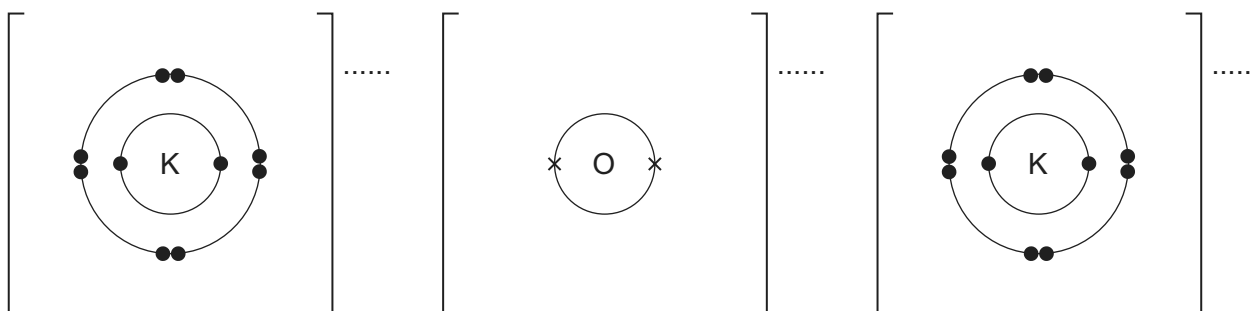


Fig. 2.1

[3]

[Total: 16]

3 The Haber process is used to manufacture ammonia.

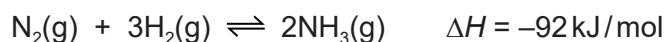
(a) State the main source of each gas used in the Haber process.

nitrogen

hydrogen

[2]

(b) The equation for the Haber process is shown.



The reaction is reversible. The forward reaction is exothermic.

(i) State what is meant by the symbol ΔH .

..... [1]

(ii) ΔH for the forward reaction is -92 kJ/mol .

State why this value shows that the forward reaction is exothermic.

..... [1]

(iii) State the typical conditions and name the catalyst used in the Haber process.

temperature °C

pressure kPa

catalyst

[3]

(iv) Complete Table 3.1 to show the effect, if any, when the typical conditions in the Haber process are changed. Use only the words **increases**, **decreases** or **no change**.

Table 3.1

change to typical conditions	effect on the rate of the forward reaction	effect on the concentration of $\text{NH}_3(\text{g})$ at equilibrium
temperature increases	increases	
pressure decreases		
no catalyst	decreases	

[4]

- (v) Explain in terms of collision theory why increasing the temperature increases the rate of the reaction.

.....

.....

.....

.....

..... [3]

- (c) Ammonia reacts with an acid to form ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$.

- (i) State the formula of the acid used.

..... [1]

- (ii) State **one** use of ammonium sulfate.

..... [1]

- (iii) Calculate the percentage composition by mass of nitrogen in $(\text{NH}_4)_2\text{SO}_4$.

percentage of nitrogen = % [2]

[Total: 18]

4 Copper is element 29 in the Periodic Table.

(a) Brass contains copper.

(i) Name the other metal in brass.

..... [1]

(ii) State the term given to a mixture of a metal with another element.

..... [1]

(b) Copper can be stretched into wires. Copper wires conduct electricity.

(i) Name the property of metals which means that they can be stretched into wires.

..... [1]

(ii) Name the particles responsible for the conduction of electricity in solid copper.

..... [1]

(c) Copper is a transition element.

Some physical and chemical properties of transition elements are shown.

physical properties:

- high density
- high strength

chemical properties:

- form coloured compounds
- have ions with variable oxidation numbers

(i) State one **other** physical property of transition elements.

..... [1]

(ii) State one **other** chemical property of transition elements.

..... [1]

(d) Hydrated copper(II) sulfate is a coloured compound. It exists as hydrated crystals which contain water molecules.

(i) State the term given to water molecules present in hydrated crystals.

..... [1]

(ii) State the colour of hydrated copper(II) sulfate crystals.

..... [1]

(iii) Write the formula of hydrated copper(II) sulfate.

..... [2]

(e) Copper(II) oxide is formed when copper(II) nitrate, $\text{Cu}(\text{NO}_3)_2$, is heated.



(i) State the class of oxide to which copper(II) oxide belongs.

..... [1]

(ii) State the meaning of the Roman numeral (II) in the name copper(II) oxide.

..... [1]

(iii) 0.0200 moles of $\text{Cu}(\text{NO}_3)_2$ is heated.

Calculate the mass of 0.0200 moles of $\text{Cu}(\text{NO}_3)_2$.

mass = g [2]

(iv) Calculate the **total** volume of gas, in dm^3 at r.t.p., produced when 0.0200 moles of $\text{Cu}(\text{NO}_3)_2$ is heated.

volume = dm^3 [2]

(v) Powdered aluminium reduces copper(II) oxide.

Write the symbol equation for this reaction.

..... [2]

[Total: 18]

- 5 Propane, propene, propan-1-ol and propanoic acid are members of different homologous series. Molecules of these substances contain three carbon atoms.

(a) Explain why members of a homologous series have similar chemical properties.

..... [1]

(b) Name the homologous series to which propanoic acid belongs.

..... [1]

(c) State the general formula of the homologous series to which propanoic acid belongs.

..... [1]

(d) Propan-1-ol has an unbranched isomer.

- Name this isomer.

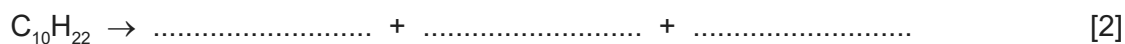
.....

- Draw the displayed formula of this isomer.

[2]

(e) Propane and propene can be manufactured by heating decane, $C_{10}H_{22}$, in the presence of a catalyst. One other product is formed.

(i) Complete the equation for this reaction.



(ii) Name this manufacturing process.

..... [1]

(f) Propene forms a polymer named poly(propene).

(i) Draw the displayed formula of a section of poly(propene) showing **three** repeat units.

[2]

(ii) State the type of polymerisation that occurs when propene forms poly(propene).

..... [1]

(g) Propanoic acid reacts with aqueous sodium carbonate to form a salt.

(i) Suggest the name of the salt formed.

..... [1]

(ii) Suggest the formula of the anion in this salt.

..... [1]

(h) Propanoic acid forms an ester when it reacts with ethanol in the presence of a catalyst.

(i) Suggest a suitable catalyst.

..... [1]

(ii) Name the ester formed.

..... [1]

(iii) Draw the displayed formula of this ester.

[2]

[Total: 17]

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The Periodic Table of Elements

Group																																				
I	II											III	IV	V	VI	VII	VIII																			
		1 H hydrogen 1																																		
		2 He helium 4																																		
3	Li lithium 7	4	Be beryllium 9	atomic number atomic symbol name relative atomic mass										5	B boron 11	6	C carbon 12	7	N nitrogen 14	8	O oxygen 16	9	F fluorine 19	10	Ne neon 20											
11	Na sodium 23	12	Mg magnesium 24											13	Al aluminium 27	14	Si silicon 28	15	P phosphorus 31	16	S sulfur 32	17	Cl chlorine 35.5	18	Ar argon 40											
19	K potassium 39	20	Ca calcium 40	21	Sc scandium 45	22	Ti titanium 48	23	V vanadium 51	24	Cr chromium 52	25	Mn manganese 55	26	Fe iron 56	27	Co cobalt 59	28	Ni nickel 59	29	Cu copper 64	30	Zn zinc 65	31	Ga gallium 70	32	Ge germanium 73	33	As arsenic 75	34	Se selenium 79	35	Br bromine 80	36	Kr krypton 84	
37	Rb rubidium 85	38	Sr strontium 88	39	Y yttrium 89	40	Zr zirconium 91	41	Nb niobium 93	42	Mo molybdenum 96	43	Tc technetium —	44	Ru ruthenium 101	45	Rh rhodium 103	46	Pd palladium 106	47	Ag silver 108	48	Cd cadmium 112	49	In indium 115	50	Sn tin 119	51	Sb antimony 122	52	Te tellurium 128	53	I iodine 127	54	Xe xenon 131	
55	Cs caesium 133	56	Ba barium 137	57–71 lanthanoids			72	Hf hafnium 178	73	Ta tantalum 181	74	W tungsten 184	75	Re rhenium 186	76	Os osmium 190	77	Ir iridium 192	78	Pt platinum 195	79	Au gold 197	80	Hg mercury 201	81	Tl thallium 204	82	Pb lead 207	83	Bi bismuth 209	84	Po polonium —	85	At astatine —	86	Rn radon —
87	Fr francium —	88	Ra radium —	89–103 actinoids			104	Rf rutherfordium —	105	Db dubnium —	106	Sg seaborgium —	107	Bh bohrium —	108	Hs hassium —	109	Mt meitnerium —	110	Ds darmstadtium —	111	Rg roentgenium —	112	Cn copernicium —	113	Nh nihonium —	114	Fl flerovium —	115	Mc moscovium —	116	Lv livermorium —	117	Ts tennessine —	118	Og oganesson —